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Best Practices for Building a Modern and Resilient Supply Chain Framework in the Oil and Gas Industry

Akash Sharma, Enverus, Houston, Texas, USA; Dan McDonald, Enverus, Tulsa, Oklahoma, USA

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Abstract

The oil and gas industry operates in a complex and challenging environment with significant supply chain management risks. This paper aims to provide a comprehensive review of these risks and proposes strategies to create a best-in-class supply chain practice using Industry 4.0 technologies. With energy prices being highly volatile and the macroeconomic situation created by COVID-19, having a resilient and robust supply chain practice is essential for oil and gas operators. The paper emphasizes the need for the energy industry to shift its view of the supply chain from a tactical, reactive organization to one of strategic value.

To establish a best-in-class supply chain practice, the paper discusses the integration of various Industry 4.0 technologies, such as artificial intelligence, robotics, and automated systems. The authors suggest leveraging tools such as spend analytics and data modeling to make informed decisions and automate dispatch processes, leading to positive fiscal and ESG impacts. They also emphasize the critical role of automation of the payables process and real-time visibility of validated invoice line-item cost detail data to achieve a should-cost modeling environment. The authors propose a manage-by-exception model that focuses on operations, value creation, process improvement, and automation of redundant support functions to mitigate risk and provide a competitive advantage for the entire enterprise. The paper presents novel and additive ideas for modernizing the oil and gas supply chain.

Background – The Why

In the contemporary oil and gas industry landscape, developing a best-in-class SCM system transcends strategic implications; it is an operational necessity. The risks of falling behind this transformation curve are far-reaching, potentially leading to operational bottlenecks, escalated costs, and competitive disadvantage. On the contrary, firms that adapt and adopt new SCM practices will enjoy enhanced operational efficiencies, cost advantages, and improved strategic positioning.

Theme I: The Indispensable Shift towards Strategic Supply Chain Management

In an era characterized by unstable market dynamics, such as the fluctuating West Texas Intermediate (WTI) prices and the disruptions instigated by the COVID-19 pandemic, the necessity for a resilient Supply Chain Management (SCM) system in the oil and gas industry has been brought to the forefront. As these

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variables induce an escalating demand for strategic SCM practices, companies are compelled to pivot from traditionally tactical approaches to more sustainable and strategic ones.

Other industries with tighter operating margins were swift in identifying the strategic value of SCM as a tool for resource optimization and cost management. Conversely, the energy industry has been slower in acknowledging this value. The urgency for a robust, strategic supply chain has become apparent in light of recent market volatility and the global impacts of COVID-19. Companies that neglect the development of sustainable SCM practices expose themselves to the risks of market volatility and global crises, potentially endangering both their operational and financial stability.

Theme II: Harnessing Industry 4.0 for a Sustainable Supply Chain

Industry 4.0 or the Fourth Industrial Revolution ushers in a slew of technological advancements, such as automation, data exchange, and the Internet of Things (IoT), enabling companies to explore sustainable and efficient SCM practices. Embracing these technologies can bolster supply chain efficiency, resilience, and strategic value, leading to sustainable digital transformation.

The advantages of a successful digital transformation are substantial, offering enhanced operational control, enriched insights, and increased transparency across commercial support functions. However, the path to digital transformation is rife with challenges, from managing change and enhancing communication to understanding existing systems. Companies that hesitate to undertake this transformation risk operational stagnation, diminished transparency, and compromised competitiveness.

Theme III: Capitalizing on Digitization and the Dire Consequences of Inefficiency

The energy industry stands at the brink of a disruption, with the potential for automation and digitization of supply chain operations creating opportunities for improved sustainability and profitability. Early adopters are more likely to navigate their digital transformations successfully, whereas those reluctant to change risk lagging behind in operational efficiency and profit optimization.

Moreover, failure to integrate and automate all business support functions on a unified platform can prevent the creation of a value-centric ecosystem, restricting real-time visibility crucial for adaptive planning in a volatile industry. Efficient and automated systems, conversely, can enhance industry responsiveness and foster sustainable profitability.

Theme IV: Integrating ESG Goals into Supply Chain Management

As companies strive to meet the escalating demands for sustainability, the integration of Environmental, Social, and Governance (ESG) considerations into SCM has become a critical element. In addition to carbon offset purchases, corporations need to make substantive strides towards reducing their carbon footprints and increasing transparency with investors. Companies that can meet these ESG goals and emission reduction targets have the potential to trade at higher valuations, ultimately contributing to improved financial performance and investment attractiveness.

Methodology – The How

Establishing a "Best-In-Class" Supply Chain organization is often a complex endeavor, with no universal formula for success. However, we aim to distill several key concepts in this section that may guide an oil and gas company from its existing ecosystem and strategy towards a considerably more resilient and advanced one. No matter where a company currently positions itself on the spectrum from nascent to advanced supply chain operations, the adoption of some, or ideally all, of these concepts promises to usher in substantial enhancements in their supply chain strategy. As previously discussed, these improvements have the potential to unlock significant value, thereby bolstering the overall operational efficiency of the organization.

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Concept I: Digitalization and Automation in Supply Chain Management (SCM)

The impact of digital transformation in Supply Chain Management (SCM) necessitates a deep understanding of two integral aspects: digitalization and automation. These transformative elements can significantly boost the supply chain's efficiency and equip organizations with the agility required to adapt to variable market conditions and consumer demands. By facilitating seamless transactions and information flow, they pave the way for a robust, resilient, and dynamic supply chain.

Digitalization in the context of SCM refers to the strategic employment of digital technologies to convert information into a digital format. This process is primarily concerned with harnessing the power of data, transforming it into actionable insights, and enabling the automation of supply chain processes. At the heart of digitalization is the challenge of ensuring data integrity and accessibility. High-quality data is the lifeblood of digital supply chains, providing the foundation for functions such as spend analytics and data modeling. These tools can yield invaluable insights into the operational efficiency of the supply chain, offering a clear picture of where improvements can be made. Digitalization underpins strategic decision-making and risk mitigation, making it a cornerstone of effective SCM.

The concept of automation is deeply entwined with digitalization. It is concerned with the use of technologies like Artificial Intelligence (AI), Machine Learning (ML), and robotics to expedite and streamline supply chain operations. From warehouse management to delivery, automation can bring about revolutionary changes, enhancing efficiency, reducing errors, and thereby bolstering both fiscal and Environmental, Social, and Governance (ESG) outcomes. By transforming specific processes such as dispatch through ticketing, accounts payable, and invoicing, automation can generate considerable efficiency gains, not to mention cost savings. By automating routine tasks, staff can focus more on strategic areas, thereby further driving organizational growth. In the energy industry, these technological shifts are not just advantageous—they're imperative for long-term sustainability and competitiveness.

Looking specifically at the payables process part of supply chain management, automation presents itself as a compelling strategic move. Particularly for those transactions that involve low-value invoices, the benefits of automation are immense. Automating this aspect of the process frees up valuable time for various internal teams, including engineering, accounting, and senior management, and allows them to focus on tasks that carry a higher value and strategic relevance. In traditional models, these teams often get mired in routine transactional tasks, such as price validation, scope and quantity verification, and coding validation. Automating these tasks could streamline operations significantly, ensuring faster processing times, fewer errors, and overall efficiency enhancement.

While the upfront investment in technology might seem substantial, the long-term benefits offered by automation certainly outweigh the initial costs. When payables processes are automated, companies can redirect their human capital towards high-value tasks such as cost analysis, financial forecasting, and budget control. Automation could also help in eliminating manual errors, improving compliance, and increasing transparency, making the entire payables process more effective and manageable. Thus, an investment in automating the payables process is essentially an investment in future operational efficiency, cost-effectiveness, and value creation.

Concept II: Emphasis on a ‘Manage-by-Exception’ Model

The ‘Manage-by-Exception’ model stands as a cornerstone in modern supply chain management, creating a shift in business operations towards more strategic and value-added activities. By automating routine tasks and inviting human intervention only when anomalies arise or predefined parameters are breached, this model effectively optimizes the utilization of resources. Exceptions can include a myriad of issues, such as supply chain disruptions, inventory discrepancies, delivery delays, or quality concerns, to which the organization must swiftly respond to maintain operational flow.

The benefits of the ‘manage-by-exception’ model are numerous, offering a rapid response mechanism to potential problems, thus reducing their overall impact on operations and leading to cost savings and

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improved service levels. Furthermore, it facilitates better resource allocation by reducing the effort expended on mundane tasks, allowing skilled staff to focus on areas where their expertise is truly needed. This redirection of resources can result in improved job satisfaction, productivity, and ultimately, bolster an organization's growth and profitability.

However, this model's implementation isn't without its trials. It demands substantial investment in technology, data analytics, and training, along with a significant shift in organizational culture, underlining trust and employee empowerment. A sturdy source-to-pay platform can be a game-changer, capable of automating routine operations, providing real-time insights, and swiftly pinpointing anomalies, thereby enabling effective supply chain risk management and consistent customer value delivery.

Concept III: Cost Modeling and Data-Driven Decision Making

Adopting a robust cost modeling approach and grounding it firmly within the framework of real-time, dependable data is integral to the success of any modern supply chain strategy. Cost modeling is a proactive measure that evaluates the constituent costs of a product or service, giving stakeholders an estimate or a benchmark for what it 'should cost'. This model provides an in-depth understanding of costs, enabling businesses to identify areas of potential savings and ways to enhance operational efficiency. From material and labor costs to overhead and profit, cost modeling allows organizations to establish more precise, realistic budgets and set strategic pricing for products or services.

Figure 1 shows the different approaches to cost modeling adopted by various organizations in the Energy industry. Sharma et.al (2023) covered Market Factors Modeling and Integrated Cost modeling in greater detail with a Haynesville example. Effective use of cost models can have a significant impact on the ability of an organization to run resilient operations. Cost modeling isn't just about number-crunching; it's about leveraging data to generate actionable insights. The intersection of performance and risk management is a pivotal juncture in this process. By harmonizing these two components, companies can paint a comprehensive picture of their most significant value streams and identify potential risks in their supply chains. This fusion of information is what gives a cost model its predictive power and enables more accurate forecasting, strategic planning, and risk management. When performance metrics and risk indicators are compiled and analyzed together, a more holistic understanding of the supply chain is formed, facilitating better decision-making processes.

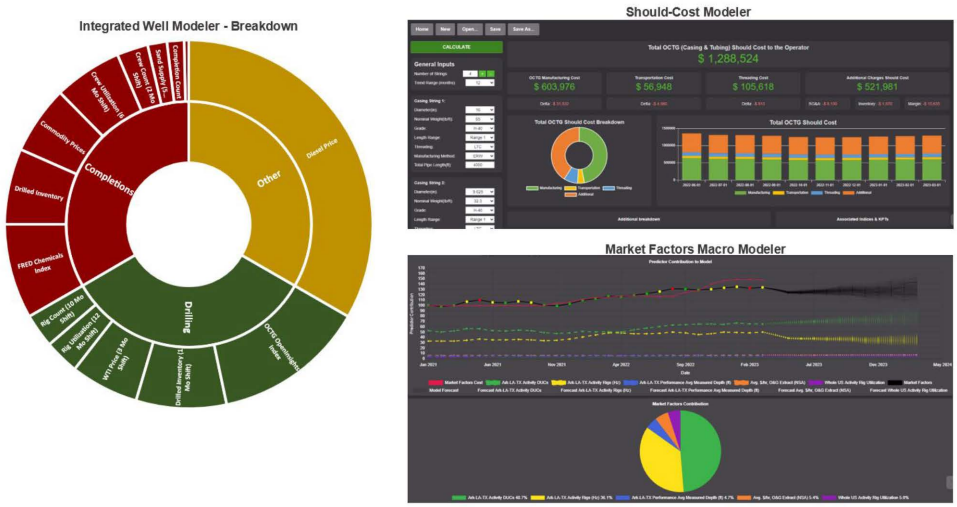

Figure 1—Different cost modeling strategies adopted by different organizations in the energy industry. Left (a) Shows an Integrated Well Cost Modeler. (b) Upper-right Shows a Should-Cost Modeler and (c) right-lower shows a Market Factors Macro Modeler.

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Data forms the core of cost modeling, and therefore, using reliable, timely data is key to developing effective models. In the age of digitalization, organizations have access to an ever-increasing volume of data, but it's crucial that this data be of high quality and easily accessible. A flawed or unreliable data set can lead to inaccurate cost models, which in turn can cause misguided strategies and financial losses. On the other hand, if used correctly, accurate data can reveal inefficiencies, bottlenecks, and opportunities for improvement. Consequently, the emphasis should be on building robust data infrastructure and cultivating a data-driven culture. A strong, data-driven cost model can not only result in cost optimization but also lead to improved supplier negotiations, enhanced quality control, and efficient resource allocation. Ultimately, it helps organizations stay competitive in a rapidly evolving market.

Concept IV: Data Integrity is King – For Analytics and Governance

The pivotal role of data integrity and accessibility in digitalized supply chain management cannot be overstressed. Essentially, it's the foundation of digital transformation in SCM, underpinning every decision, strategy, and subsequent action. For an organization to transition successfully into the digital era, it must ensure that it has a robust mechanism in place to gather accurate, reliable, and actionable data from a plethora of sources.

Data integrity refers to the accuracy and consistency of data over its entire lifecycle. High data integrity guarantees that the information is exactly as intended, free from any corruption or alteration from the original. It is the prerequisite for data reliability and the cornerstone for data-driven decision-making. This is especially crucial in SCM, as decisions made based on flawed or inconsistent data can lead to inefficiencies, decreased productivity, and potentially, financial losses.

On the other hand, data accessibility refers to how readily available and understandable data is to those who need it. With easily accessible data, organizations can make swift, informed decisions, leading to enhanced agility and resilience. Accessible data also facilitates transparency, fostering trust among stakeholders. However, managing these data aspects can be a daunting task due to the sheer volume and complexity of the information being processed. Therefore, it's essential for organizations to adopt sophisticated data management tools, and possibly, AI and machine learning technologies, to ensure data integrity and accessibility while gleaning valuable insights for strategic operations. After all, the power of digitalization in SCM lies in harnessing data to its full potential – transforming raw data into actionable intelligence for optimal operational outcomes.

Concept V: Strategic Sourcing and Category Management

Strategic sourcing and category management form an integral part of efficient supply chain management, particularly in the context of the oil and gas industry. These techniques, when bolstered by data-driven insights, can foster proactive decision-making, cost optimization, and risk mitigation - the holy grail of any robust SCM framework.

Strategic sourcing is a systematic process, allowing companies to reevaluate and optimize their purchasing activities. The focus is on improving overall business performance by understanding total spending, reducing costs, streamlining operations, and improving the quality of products and services. It's not just about obtaining the lowest price but about establishing long-term partnerships with suppliers to generate sustained value. This process, when armed with relevant, accurate data, can unearth significant opportunities for operational improvements and cost savings.

Meanwhile, category management goes a step further, helping organizations manage their supply categories as strategic business units. This practice revolves around sourcing strategically important categories that drive significant value to the business. Category management involves an end-to-end approach, right from identifying the need for a particular category to finalizing contracts with suppliers. Using a data-driven approach, category management aims at maximizing value from the supply base,

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thus promoting competitive advantage. It facilitates market intelligence and supplier insights, promoting innovation, minimizing risks, and improving speed to market.

In essence, strategic sourcing and category management, when fueled by trusted real-time data, can result in a shift from a traditionally reactive approach to a proactive one. This proactive 'management-by-exception' approach enables organizations to direct their focus on value-add activities, rather than getting mired in the mundane. This approach ensures that any deviations from the norm or areas of concern are given the necessary attention, leading to improved decision-making and efficient SCM practices. By harnessing the power of these tools, organizations can embark on a journey toward digital transformation, propelling their operations to new heights of efficiency.

Concept VI: Source-to-Pay as an Integrated Strategy

In the realm of Supply Chain Management, the source-to-pay (S2P) process stands out as a critical strategy that encompasses a gamut of activities starting from identifying and sourcing the supplier of goods or services, to the fulfillment and final payment for those services. The S2P process isn't merely a series of steps, but rather a strategic approach that can influence an organization's operational efficiency and bottom line.

The early stages of the S2P process involve identifying and engaging with suppliers who can provide the goods and services needed by an organization. This stage is highly strategic, requiring a well-considered evaluation of potential suppliers. Factors such as the supplier's capacity, quality standards, cost-effectiveness, and track record for reliability should be analyzed. Upon successful negotiation and contracting, a mutually beneficial relationship is established, with the organization securing necessary resources and the supplier gaining a stable customer.

However, the strategic value of the S2P process doesn't end with contracting. The subsequent stages involving the receipt and validation of goods or services and their final payment are equally critical. Invoice and payment automation, powered by sophisticated software solutions, not only saves time and eliminates human errors but also provides real-time visibility into expenditure. Furthermore, the utilization of analytics within the S2P process can aid in pinpointing inefficiencies and identifying areas of cost savings. In this way, the S2P process serves not only as an operational tool but also as a strategic driver of business value.

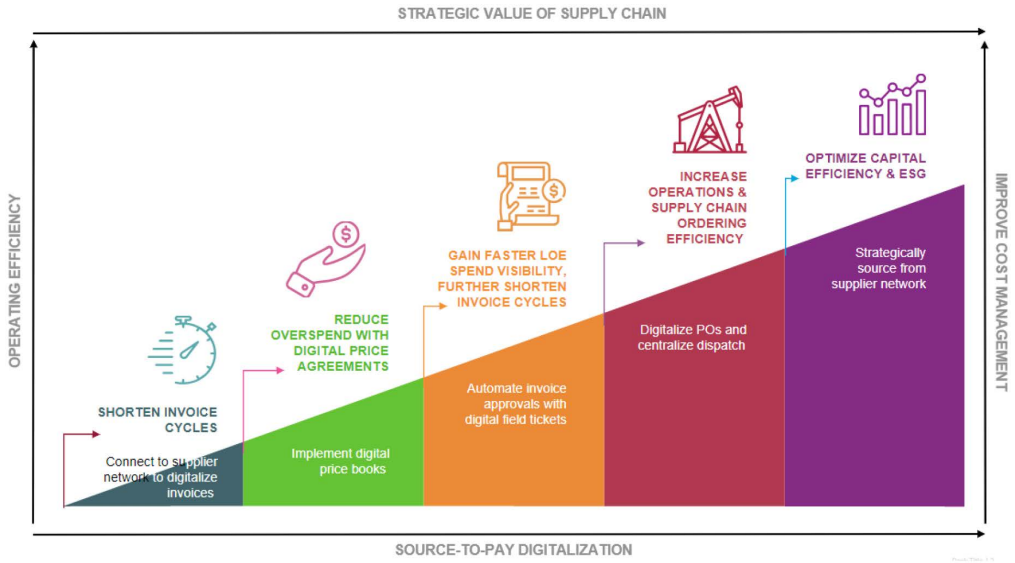

Figure 2—Incremental digitalization driving amplified efficiency in a Source-to-Pay Strategy.

When S2P is integrated with strategic sourcing, it can provide an even more powerful solution. Strategic sourcing, which focuses on leveraging purchasing power to achieve the best possible total cost of ownership

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and minimize risk, is another essential tool in the SCM toolkit. An integrated approach can ensure that sourcing decisions align with broader supply chain and business strategies, thus providing an integrated spend and risk management solution. By using data from the S2P process, strategic sourcing can be more precise, efficient, and effective. In sum, the strategic role of the source-to-pay process in supply chain management is undeniable, with its ability to drive efficiency, enhance visibility, and reduce risk.

Concept VII: The Digital Brain Ecosystem: The Human Body Analogy

The digital brain ecosystem in the oil and gas industry can be conceived as a human body where each element plays a vital role in sustaining the overall system. This interconnected network starts with the data - the food for the digital brain. Payables automation effort feeds clean, reliable, and properly coded data into the system, comparable to a healthy diet that fuels sound decision-making processes. This initiative facilitates real-time, enterprise-wide visibility of line-item cost details, standardized categorization and descriptions, ensuring that data integrity is protected and spend management is efficient.

Figure 3—Illustration of the digital brain ecosystem

An electronic delivery receipt, akin to a validation tool, maintains data integrity, confirming the scope and quantity with mobile capability. Further, this tool can audit unsupervised field labor using IoT tools such as cameras, GPS, and Geo-fencing, enhancing validation. By fully utilizing payables automation, invoices can essentially be phased out, recognizing costs upon completion of work and approval of the electronic delivery receipt, rather than 30-90 days later, thereby eliminating the accrual process.

With the ‘pay’ part of the source-to-pay process automated, focus shifts to the ‘sourcing’ aspect. The data can guide an intelligent sourcing tool designed for materials procurement and complex services, enabling an accurate, value-based comparison of offerings. This can be thought of as the ‘hands and feet’ of the body, moving or acting based on instructions from the ‘brain’, or in this context, the digital brain informed by data, analytics, and market intelligence.

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Your data set could also power your strategic sourcing and category management tools. Effective category management, similar to a strategic approach in managing a group of products or services, relies on analyzing market trends, understanding customer needs, and developing strategies to optimize procurement and pricing. Your commercial documentation can flip into a pricing validation tool, much like the body's immune system, ensuring you are being charged as per your negotiations and affirming control over your spend.

Picture all these tools feeding into a management-by-exception dashboard with multi-level visibility and drill-down capabilities. This ‘vision’ could facilitate supplier network visibility, supplier onboarding, risk vetting, and CapEx/OpEx budget tracking. Eventually, this could enable should-cost modeling and scenario planning, while easy access to market intelligence would provide insights on how external events could impact your supply chain.

In an increasingly agile world, you need a resilient supply chain that can withstand the unexpected. A future-facing organization would leverage this ecosystem to replace tactical paper-pushing with strategic, data-informed decision-making. The effectiveness of this digital ecosystem depends on the quality of the data input; inferior data will invariably lead to inferior insights, which in turn will impact decision-making. Just like a healthy body relies on a balanced diet, a robust digital brain ecosystem is fundamentally contingent on the quality of the data it processes.

Conclusions

In conclusion, this paper has explored the critical need for the oil and gas industry to develop a best-in-class supply chain management (SCM) practice in the face of complex and volatile market dynamics. By embracing Industry 4.0 technologies and shifting from a tactical to a strategic approach, companies can achieve enhanced operational efficiencies, cost optimization, and improved strategic positioning.

The integration of various Industry 4.0 technologies, such as artificial intelligence, robotics, and automated systems, is essential for achieving a resilient and advanced supply chain practice. Tools like spend analytics and data modeling enable informed decision-making and automation of processes, leading to positive fiscal and Environmental, Social, and Governance (ESG) impacts. Automation of the payables process and real-time visibility of validated invoice line-item cost detail data can drive efficiency and establish a should-cost modeling environment.

The paper also highlights the importance of adopting a "manage-by-exception" model, focusing on operations, value creation, process improvement, and automation of redundant support functions. This approach optimizes resource utilization and enables rapid response to supply chain disruptions, leading to cost savings and improved service levels.

Additionally, the strategic use of cost modeling and data-driven decision making allows organizations to identify areas of potential savings, enhance operational efficiency, and mitigate risks. Data integrity and accessibility are critical for successful digital transformation, empowering organizations to harness the full potential of data for actionable insights and optimal decision-making.

Furthermore, integrating Environmental, Social, and Governance (ESG) considerations into supply chain management is crucial in today's sustainability-focused landscape. Meeting ESG goals and reducing carbon footprints contribute to improved financial performance and investment attractiveness.

The concept of a source-to-pay (S2P) strategy emerges as a key enabler for a best-in-class supply chain practice. Integrating strategic sourcing, category management, and automation in the S2P process enhances operational efficiency, cost optimization, and risk management. The S2P process serves as a critical driver of business value, facilitating efficient supplier management, real-time visibility, and improved decision-making.

Finally, the paper presents the Digital Brain ecosystem, an analogy comparing the interconnected elements of supply chain management to the various components of the human body. Data acts as the fuel for the digital brain, driving informed decision-making, while automation and analytics tools play vital roles

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in streamlining processes and providing insights. The effectiveness of this ecosystem relies on the quality of the data input, akin to a healthy body depending on a balanced diet.

By adopting the concepts and strategies discussed in this paper, the oil and gas industry can modernize its supply chain, achieve operational excellence, and gain a competitive advantage in an increasingly dynamic and demanding market. The journey toward a best-in-class supply chain practice is a transformative one, requiring organizational commitment, investment in technology, and a data-driven culture. However, the potential rewards in terms of efficiency, cost savings, risk mitigation, and sustainability make it a worthwhile endeavor for companies seeking long-term success.

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